Principal AI Infrastructure Guide to Qwen3.5:0.8B Deployment and Prompt Engineering

The landscape of edge-based artificial intelligence was fundamentally altered on March 2, 2026, with Alibaba Cloud's release of the Qwen3.5 Small Model Series. At the center of this release is the Qwen3.5:0.8B variant, an ultra-compact multimodal foundation model that serves as the definitive proof-of-concept for the "More Intelligence, Less Compute" philosophy.[1] As the smallest entry in the Qwen3.5 family, which scales up to 397B parameters, the 0.8B model is not a mere distillation or a shrunken afterthought of its flagship counterparts.[4] Instead, it represents a ground-up re-engineering of the Small Language Model (SLM) category, utilizing a hybrid architecture that fuses linear attention mechanisms with Gated Delta Networks to overcome the memory and compute bottlenecks that have historically limited sub-1B models.[4] This report serves as a comprehensive integration guide for Principal AI Infrastructure Engineers and SLM Deployment Specialists, detailing the nuances of the model's native multimodality, its revolutionary thinking mode, and the empirical complexities of its few-shot performance.

Executive Summary: The Intelligence Density Breakthrough

The primary innovation defining Qwen3.5-0.8B is its intelligence density, a metric that describes the model's ability to achieve frontier-level reasoning scores relative to its parameter count.[2] While prior generations like Qwen2.5 and the early Qwen3 iterations focused on optimizing dense transformer blocks, Qwen3.5-0.8B is built on the "Qwen3-Next" foundation.[8] This architecture introduces Gated Delta Networks (GDN), which replace standard softmax attention in 75% of the model's layers.[4] By utilizing linear attention for routine sequence processing and reserving full attention for complex relational tasks, the 0.8B variant maintains a context window of 262,144 tokens natively, a capability previously restricted to enterprise-scale LLMs.[1]

Infrastructure specialists should recognize that Qwen3.5-0.8B is the first model in its class to integrate native multimodal capabilities via early fusion training.[6] Rather than utilizing a separate vision encoder attached as a "bolted-on" adapter, the model processes text, images, and video frames within a unified latent space from the first stage of training.[12] This structural integration allows the model to excel in "pixel-level positioning" and "visual agentic workflows," such as autonomously navigating smartphone UIs or performing high-accuracy OCR on dense financial documents.[6] Benchmark results indicate that the 0.8B model achieves an MMLU-Pro score of

42.3 in its thinking mode, effectively surpassing the performance of the prior Qwen3:30B model on several key reasoning benchmarks while requiring only 2GB of VRAM for full-precision inference.[5]

The model's release was accompanied by significant organizational shifts within Alibaba's Tongyi Lab. The departure of key technical leads, including Lin Junyang, immediately following the rollout suggests a strategic pivot from pure research to commercial implementation.[17] For infrastructure engineers, this highlights the 0.8B model's importance as a stable, open-source anchor point for edge deployment, licensed under Apache 2.0 to ensure long-term viability and community-driven optimization.[1]

Feature	Qwen 2.5 (0.5B)	Qwen 3.0 (0.6B)	Qwen 3.5 (0.8B)
Architecture	Dense Transformer	Hybrid MoE (Experimental)	Gated DeltaNet + sparse MoE
Context Window	32,768	128,000	262,144 (Native)
Multimodality	Adapter-based	Adapter-based	Natively fused early-fusion
Thinking Mode	Not Supported	Experimental	Fully Integrated Hybrid
Vocab Size	151,643	151,936	248,320
Max Context	32k	128k	1,010,000 (via YaRN)

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Hybrid Thinking Mode: Syntax and Logics of Deliberation

The integration of a Hybrid Thinking Mode is perhaps the most distinctive feature of the Qwen3.5 generation. This mode allows the model to generate internal Chain-of-Thought (CoT) sequences, providing a glimpse into its reasoning process and significantly improving accuracy on multi-step logical tasks.[20] For the 0.8B model, this capability is a dual-edged sword: while it boosts the model's effective intelligence, its limited parameter count makes it prone to repetitive thinking loops if not managed through precise hyperparameter tuning and prompt

structure.[14]

The Reasoning Syntax Guide

The Qwen3.5:0.8B model utilizes a structured response format to separate internal deliberation from the final output. When Thinking Mode is engaged, the model produces a thought block wrapped in specific tags: <think>...</think>.[14] This is followed by the standard assistant response.[20] Unlike the previous Qwen3 generation, which supported a "soft switch" via /think or /no_think commands in the user prompt, Qwen3.5-0.8B prioritizes the use of a hard switch via the enable_thinking parameter at the API or inference server level.[26]

Infrastructure engineers must configure their parsing engines to handle this structured output. If the reasoning content is not extracted by the server (e.g., using vLLM's qwen3 parser), the final user will see the raw deliberation as part of the chat response.[22] This is critical for agentic workflows where the reasoning process might contain confidential information or "intermediate hallucinations" that the model self-corrects before its final answer.[16]

Thinking vs. Non-Thinking Trade-offs

The decision to enable thinking for a 0.8B model is fundamentally a trade-off between latency and reliability. Non-thinking mode, also referred to as "Instruct" or "Instant" mode, delivers the direct responses typical of standard SLMs.[20] In this state, the model relies on its pre-trained weights to predict the next token immediately.[21]

In Thinking Mode, the model allocates a "computation budget" to internal reasoning.[21] This process is highly beneficial for mathematical reasoning, coding, and complex document interpretation.[20] For instance, on the MMLU-Pro benchmark, the 0.8B variant's score jumps from 29.7 (Non-Thinking) to 42.3 (Thinking).[14] However, this improvement comes at the cost of a significantly higher output token count and increased time-to-first-complete-token (TTFCT).[30]

Metric	Non-Thinking Mode	Thinking Mode (Thinking)	Source
MMLU-Pro	29.7	42.3	
C-Eval	46.4	50.5	
MMMLU	34.1	44.3	
IFEval	52.1	44.0 (See Note)	

Note: The slight decrease in IFEval during thinking mode is a known artifact of current SLM architectures, where extensive internal deliberation can occasionally lead the model to deviate from strict formatting constraints.[14]

Multimodal Vision Prompting Structure

Qwen3.5-0.8B represents a paradigm shift in visual-language modeling (VLM) for edge devices. By implementing an early-fusion backbone trained on trillions of diverse multimodal tokens, the model achieves spatial intelligence capabilities that were previously exclusively available in models 10-20 times its size.[2] This allows the 0.8B model to handle sophisticated tasks such as identifying defects in circuit boards, extracting structured data from dense service agreements, and reverse-engineering programmatic logic from gameplay footage.[16]

Image and Video Input Protocols

The model adheres to a unified multimodal interface, typically accessed via an OpenAI-compatible API.[14] Images and videos are passed as part of the message content list, allowing for interleaved text and media.[14]

The technical specification for media input is highly flexible, supporting JPEG, PNG, and WebP for images, and MP4 or WebM for videos.[36] For infrastructure engineers running local servers, the following JSON structure is the functional standard for a multimodal request:

JSON

```
"messages":
  }
]
```

When processing video, the model's 3D convolution layers capture temporal dynamics across frames.[16] Official documentation recommends a sampling rate of 2 FPS (frames per second) for most understanding tasks, though this can be scaled up to 8 FPS for high-precision element tracking in UI navigation.[10] The model supports up to 60 seconds of video natively in its 0.8B form, provided the GPU offloading is managed correctly to handle the media embedding size.[10]

Spatial Intelligence and Pixel-Level Grounding

The most advanced applications of Qwen3.5-0.8B involve its spatial intelligence features.[6] The model can return coordinate-based data, enabling "visual agentic control" where it identifies

clickable elements on a screen and generates the corresponding code to interact with them.[6] In document recognition benchmarks, it breaks the 90% threshold in OmniDocBench, outperforming larger rivals like GPT-5.2 and Claude Opus 4.5 in specific document extraction tasks.[37]

Principal AI Engineers should leverage the model's "MathVision" capability, which utilizes a specific fixed prompt structure to solve geometry and chart-based mathematical problems.[14] By including the instruction "Please reason step by step, and put your final answer within \boxed{}," engineers can trigger the model's highest accuracy reasoning path.[14]

Vision Benchmark	Qwen3.5-0.8B (Thinking)	Qwen3-VL-2B	GPT-5-Nano	Source
MMMU	49.0	61.4	45.2	
Mathvista	62.2	73.6	58.6	
RealWorldQA	63.4	69.5	61.6	
MathVision	78.9	74.6	62.2	

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The Few-Shot Paradox: Scaling Logic vs. Collapse

A critical discovery for SLM deployment specialists is the "Few-Shot Paradox" specifically observed in the Qwen3.5:0.8B variant.[22] Traditionally, prompt engineering best practices dictate that adding examples to a prompt (few-shotting) increases the model's ability to follow complex formatting and reasoning logic. However, empirical testing on the 0.8B model reveals a stark divergence based on task complexity.[22]

Classification: Successful Scaling

On narrow, structured tasks like classification, the 0.8B model behaves like its larger counterparts, exhibiting a traditional learning curve.[22] Zero-shot performance on intent detection or sentiment analysis often starts around 60%, but scales cleanly to 100% when 8-shot examples are provided via TF-IDF selection.[22] This suggests that the model's internal representations can effectively absorb the "pattern" of simple classification from contextual examples.

Code and Math: The Collapse Phenomenon

The paradox emerges in high-complexity domains such as code fixing and mathematical reasoning.[22] In these scenarios, the 0.8B model often performs *best* in a zero-shot environment.[22] Community benchmarks show the model achieving 67% accuracy on zero-shot code bug-fixes, but dropping instantly to 33% the moment a single example is added to the prompt.[22]

The technical reason for this collapse is rooted in the model's limited parameter depth and residual stream capacity.[22] For a sub-1B parameter model, the cognitive overhead of processing and "attending" to multi-file code examples in the prompt clutters its active memory, causing it to lose the original instruction.[16] Essentially, the model has sharp enough logic to solve the problem immediately but lacks the "working memory" to process external context without becoming distracted or over-correcting.[16]

The Few-Shot Decision Matrix

For AI engineers, this necessitates a task-specific prompting strategy. The following matrix should be used to determine shot-counts for the 0.8B model:

Task Category	Optimal Shot Count	Observed Behavior	Deployment Recommendation
Classification	4-8 Shots	Linear performance gain	Use high-shot context for production stability.
Code Fixing	0-Shot	Immediate performance collapse	Use zero-shot prompts with high precision sampling.
Summarization	1-2 Shots	Peak at 2 shots, then plateau	Use minimal examples to set output style only.
Agentic Planning	0-Shot	CoT provides better logic than examples	Rely on Thinking Mode rather than examples.

Deployment Infrastructure: vLLM, SGLang, and Ollama

Successful deployment of Qwen3.5:0.8B requires selecting a serving framework that supports its hybrid attention kernels and Multi-Token Prediction (MTP) pipeline.[14] For Principal Infrastructure Engineers, the choice of engine is determined by the desired throughput-to-latency ratio.

vLLM: High-Throughput Microservices

vLLM is the recommended framework for high-throughput serving of Qwen3.5-0.8B, particularly when deploying via Docker in cloud environments.[14] vLLM version 0.11.0 or higher is required to support the early-fusion vision features.[35] One critical optimization for infrastructure engineers is the use of --language-model-only when vision is not required, which skips loading the vision encoder and frees up nearly 40% of the model's VRAM requirement for additional KV cache capacity.[14]

Bash

```bash
# Optimized vLLM launch for 0.8B on a mobile server
vllm serve Qwen/Qwen3.5-0.8B \
 --port 8000 \
 --max-model-len 32768 \
 --gpu-memory-utilization 0.9 \
 --reasoning-parser qwen3 \
 --enable-auto-tool-choice \
 --tool-call-parser qwen3_coder \
 --speculative-config '{"method":"qwen3_next_mtp","num_speculative_tokens":2}'
```

The addition of the MTP speculative configuration allows the engine to predict multiple tokens per forward pass, significantly reducing per-token latency for longer reasoning chains.[14]

Ollama: Local Prototyping and Edge Privacy

Ollama provides a streamlined interface for developers running the model on local hardware.[40] The official qwen3.5:0.8b library entry is a 1GB download in Q8_0 format, making it ideal for integration into background system services on laptops and smartphones.[2]

To enable Thinking Mode in Ollama, a custom Modelfile must be created to override the default

non-thinking behavior.[44] Community engineers have discovered that injecting an empty <think></think> tag at the start of the assistant response can "force" the model into its deliberation state even if the API parameter is not explicitly supported by the client application.[26]

Hugging Face: The Development Baseline

For engineers conducting research or fine-tuning experiments, the transformers library remains the primary interface.[14] Qwen3.5-0.8B requires transformers >= 4.51.0.[20] Because the model uses a causal language model with a vision encoder, it requires the AutoProcessor and Qwen3VLForConditionalGeneration classes.[46]

Python

```python
from transformers import Qwen3VLForConditionalGeneration, AutoProcessor
import torch

model = Qwen3VLForConditionalGeneration.from_pretrained(
    "Qwen/Qwen3.5-0.8B",
    torch_dtype="auto",
    device_map="auto"
)
processor = AutoProcessor.from_pretrained("Qwen/Qwen3.5-0.8B")

# Note: enable_thinking=True is now a native kwarg in apply_chat_template
```

Optimal Hyperparameters to Prevent Thinking Loops

The "infinite thinking loop" is a common failure mode for Qwen3.5-0.8B, where the model generates hundreds of tokens of deliberation that never converge on a final answer.[22] This usually manifests as the model repeatedly outputting "But wait..." or "Let me re-check that...".[26] Preventing this requires a specific balance of sampling parameters and output constraints.

Sampling Parameters Cheat Sheet

The most effective tool for preventing loops is the presence_penalty.[20] Official documentation suggests a range of 1.5 to 2.0 to discourage the model from revisiting the same reasoning steps.[26] Conversely, repetition_penalty should be kept close to 1.0; increasing it too high can cause the model to mix languages or lose grammatical coherence.[20]

Infrastructure engineers should strictly avoid greedy decoding for the 0.8B model in thinking mode, as it leads to deterministic loops where the model gets trapped in its own self-correction logic.[20] Instead, a temperature of 1.0 for general tasks and 0.6 for precise coding tasks is recommended.[14]

Deployment Scenario	Mode	Temperature	Top-P	Top-K	Min-P	Presence Penalty
General Chat	Non-Think	0.7	0.80	20	0.0	1.5
Complex Logic	Thinking	1.0	0.95	20	0.0	2.0
Precise Code	Thinking	0.6	0.95	20	0.0	0.0
Vision/VL Tasks	Non-Think	0.7	0.80	20	0.0	1.5
Vision/VL Tasks	Thinking	0.6	0.95	20	0.0	1.5
Video Summary	Thinking	1.0	0.95	20	0.0	1.5

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The "Low Reasoning Effort" Trick

For production environments where latency is critical, engineers can utilize a "Low Reasoning Effort" trick by passing specific values to the model's internal reasoning_effort parameter (often mapped to chat_template_kwargs in vLLM or llama-server).[51] Setting effort to "low" allows the model to perform a single-pass verification rather than an exhaustive deliberation, reducing output tokens by up to 70% while maintaining much of the "thinking" accuracy gain.[51]

Advanced Integration Best Practices

Transitioning from local testing to a production-grade infrastructure requires addressing several high-level engineering challenges, specifically context management and tool-calling stability.

Context-Folding and Hallucination Cascades

Despite the 262k context window, the 0.8B model is vulnerable to "hallucination cascades" in long multi-turn conversations.[16] This occurs when a small reasoning error in turn 3 is amplified by the model's own subsequent deliberation in turn 10, eventually leading the model to pursue completely meaningless plans.[16]

Principal Engineers should implement a "context-folding" strategy.[40] Once the conversation history reaches a threshold (e.g., 64k tokens), the system should prune intermediate tool responses and thought blocks, preserving only the system prompt, the original goal, and the three most recent turns.[40] This keeps the model's limited attention focused on the immediate task rather than being "drowned" in its own prior reasoning history.

The Recursive Parser Trap

A critical bug was identified in the interaction between the Qwen3.5:0.8B model's deliberation process and common tool-calling parsers.[54] When the model "thinks" about tool calling syntax within its <think> block—for instance, reasoning "I should probably use <function=search_nodes>"—the parser may identify those prose mentions as actual tool call attempts.[54] If the parse fails, the error is fed back to the model, which then reasons about the failure, mentions the syntax again, and triggers a recursive loop.[54]

The workaround is to explicitly disable reasoning during tool-intensive turns or to utilize a parser that ignores all content within the <think> tags.[54] In vLLM, the qwen3_coder parser is specifically designed to handle this, though engineers should monitor for conflicts when Multi-Token Prediction is also enabled.[54]

Speculative Decoding Architectures

One of the most effective use cases for Qwen3.5:0.8B is as a "draft model" for the flagship Qwen3.5-397B-A17B.[55] In this architecture, the 0.8B variant runs on local edge hardware or a cheaper GPU node to generate potential token sequences.[16] These sequences are then validated by the 397B model on an enterprise cluster.[16] This "speculative orchestration" can increase the flagship model's throughput by 2.75x, enabling the deployment of massive frontier models at the cost of a medium-sized cluster.[44]

Advanced Prompt Templates for Developers

The following templates are designed for high-performance agentic and multimodal

workflows, ensuring that the 0.8B model adheres to professional-grade standards.

Agentic Microservice Template (Thinking Enabled)

This structure is optimized for models running within a larger agentic ecosystem like OpenClaw or Devin 2.2.[8]

<|im_start|>system

You are a specialized agentic microservice.

Primary Goal:

Available Tools:

Operational Constraints:

1. You MUST use the block to plan your next step.
2. If the user's intent is unclear, ask clarifying questions before reasoning.
3. Once the task is complete, provide a final answer starting with "SUCCESS:".
4. Adhere to a presence penalty of 1.5 to avoid repeating thoughts.
   <|im_end|>
   <|im_start|>user

<|im_end|>

<|im_start|>assistant

Multimodal Document Extraction Template

Designed for high-accuracy extraction from slides, tables, and handwritten notes.[12]

<|im_start|>system

You are a highly accurate vision-language document interpreter.

Task: Extract financial metrics from the provided media content.

Output Format: Return only a structured JSON object.

Constraint: Do not include conversational filler in your final answer.

<|im_end|>

<|im_start|>user

<|vision_start|><|vision_end|>

Extract the following from this slide:

1. Quarter-over-Quarter (QoQ) growth percentage.

2. Net Revenue for the International segment.
3. Operating margin listed in the footer notes.
   <|im_end|>
   <|im_start|>assistant

Zero-Shot Code Debugging Template

Optimized to avoid the Few-Shot Paradox and the collapse of the residual stream.[22]

<|im_start|>system

You are a senior software engineer specializing in low-latency systems.

Goal: Identify the bug in the provided snippet and suggest a fix.

Constraint: Provide NO EXAMPLES. Reason directly from the code.

<|im_end|>

<|im_start|>user

Current Code:

Error Message:

Fix this.

<|im_end|>

<|im_start|>assistant

Hardware and System Requirements for Engineers

Managing the deployment of Qwen3.5-0.8B requires a nuanced understanding of memory allocation, particularly for high-context use cases.

Setup Type	Format	Min VRAM	CPU/System RAM	Recommended Setup	Source
Mobile/Edge	Q4_K_M	0.5 GB	8 GB	iPhone 17 Pro / Snapdrago	[8]

				n Gen 5	
Dev Workstation	Q8_0	1.0 GB	16 GB	Mac Mini M4 / RTX 3060	[8]
Prod Microservice	Full BF16	1.8 GB	32 GB	Tesla T4 / A10G	[8]
High Context (262k)	BF16	35.1 GB	64 GB	RTX A6000 / A100 (40GB)	[19]

Engineers should note that the VRAM requirement scales exponentially with the context window length. While the model weights for a Q4_K_M quant fit in just 520MB, the KV cache for a full 262k token window in 16-bit precision requires over 30GB of VRAM.[19] For edge devices, the context window should be capped at 8,192 to 16,384 tokens to preserve system stability.[19]

Conclusions and Future Outlook

The Qwen3.5:0.8B model represents the apex of current SLM technology, providing a functional bridge between local edge devices and the frontier-class capabilities of the cloud. For Principal AI Infrastructure Engineers, the model's value lies in its extreme versatility: it can act as a high-speed classifier, a capable visual agent, or a sophisticated draft model for trillion-parameter flagships.[16]

The "Few-Shot Paradox" findings underscore that as models become smaller and more specialized through architectural tricks like Gated Delta Networks, the standard prompting strategies of the LLM era must be discarded in favor of model-aware engineering.[22] Infrastructure specialists should focus on:

1. **Zero-Shot Logic for High-Complexity Tasks:** Relying on the model's internal thinking mode rather than cluttering its prompt with examples.[22]
2. **Strategic Sampling Control:** Utilizing presence penalties of 1.5+ to mitigate the looping behavior inherent in ultra-compact thinking models.[26]
3. **Unified Multimodal Orchestration:** Leveraging the early-fusion architecture for UI navigation and document analysis without the overhead of external vision towers.[6]

Despite the internal organizational changes at Alibaba, the Qwen series remains the globally

dominant open-source lineage, with over 200,000 derivative models on Hugging Face as of early 2026.[17] The 0.8B model is a cornerstone of this ecosystem, enabling a future where "vibe coding," on-device translation, and visual workflows are no longer restricted by server availability or API costs, but are limited only by the engineer's ability to orchestrate the "Impressive Intelligence Density" of this compact powerhouse.[2]

Works cited

1. Daily AI model release digest - Scouts by Yutori, accessed on March 7, 2026, https://scouts.yutori.com/03d3d7ee-5a2b-4314-9124-b9ca8b2a3c39
2. Alibaba Releases Open-Source Qwen3.5 Small Models for Edge Devices - MLQ.ai, accessed on March 7, 2026, https://mlq.ai/news/alibaba-releases-open-source-qwen35-small-models-for-edge-devices/
3. Tech Brief (March 4): Alibaba Launches Small AI Models Focused on Efficiency, accessed on March 7, 2026, https://www.caixinglobal.com/2026-03-04/tech-brief-march-4-alibaba-launches-small-ai-models-focused-on-efficiency-102419438.html
4. The Compression Problem: How Qwen 3.5 Small Rewrites the Rules of Edge AI - Medium, accessed on March 7, 2026, https://medium.com/@AdithyaGiridharan/the-compression-problem-how-qwen-3-5-small-rewrites-the-rules-of-edge-ai-3062e8a9d5e8
5. Qwen 3.5: Scaling Intelligence in Compact Models - HowAIWorks.ai, accessed on March 7, 2026, https://howaiworks.ai/blog/alibaba-qwen-3-5-compact-models-announcement
6. Qwen3.5-0.8B-Base - 模型详情页, accessed on March 7, 2026, https://modelscope.cn/models/Qwen/Qwen3.5-0.8B-Base
7. Elon Musk praises Alibaba's Qwen model for its "impressive intelligence density" | KuCoin, accessed on March 7, 2026, https://www.kucoin.com/news/flash/elon-musk-praises-alibaba-s-qwen-model-for-impressive-intelligence-density
8. Qwen 3.5 Developer Guide: Benchmarks, Architecture & Integration | Lushbinary, accessed on March 7, 2026, https://www.lushbinary.com/blog/qwen-3-5-developer-guide-benchmarks-architecture-integration-2026/
9. The convergence between local and cloud AI models is happening faster than most people think - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rld6rh/the_convergence_between_local_and_cloud_ai_models/
10. Qwen 3.5 Developer Guide: 397B MoE Model with Visual Agents, API & Self-Hosting (2026), accessed on March 7, 2026, https://www.nxcode.io/resources/news/qwen-3-5-developer-guide-api-visual-agents-2026
11. Qwen/Qwen3.5-0.8B-Base - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3.5-0.8B-Base

12. Qwen 3.5 Explained: Architecture, Upgrades Over Qwen 3, Benchmarks, and Real-World Use Cases - Medium, accessed on March 7, 2026, https://medium.com/data-science-in-your-pocket/qwen-3-5-explained-architecture-upgrades-over-qwen-3-benchmarks-and-real-world-use-cases-af38b01e9888
13. QwenLM | Qwen 3.5 - Kaggle, accessed on March 7, 2026, https://www.kaggle.com/models/qwen-lm/qwen-3-5
14. Qwen/Qwen3.5-0.8B - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3.5-0.8B
15. Alibaba just released Qwen 3.5 Small models: a family of 0.8B to 9B parameters built for on-device applications - MarkTechPost, accessed on March 7, 2026, https://www.marktechpost.com/2026/03/02/alibaba-just-released-qwen-3-5-small-models-a-family-of-0-8b-to-9b-parameters-built-for-on-device-applications/
16. Alibaba's Qwen3.5 Small Models: 0.8B Parameters Can Process Video, The Edge AI Era Has Arrived | by AI Engineering - Medium, accessed on March 7, 2026, https://medium.com/@ai-engineering-trend/alibabas-qwen3-5-small-models-0-8b-parameters-can-process-video-the-edge-ai-era-has-arrived-65226d706133
17. Shortly after Jack Ma concluded the AI mobilization meeting, the 'key figure' behind Qwen left., accessed on March 7, 2026, https://news.futunn.com/en/post/69594961/shortly-after-jack-ma-concluded-the-ai-mobilization-meeting-the
18. Alibaba Qwen team's lead researcher and other key members suddenly resign, accessed on March 7, 2026, https://gigazine.net/gsc_news/en/20260305-alibaba-step-down/
19. Qwen3.5-0.8B: Specifications and GPU VRAM Requirements - ApX Machine Learning, accessed on March 7, 2026, https://apxml.com/models/qwen35-08b
20. Qwen/Qwen3-8B - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3-8B
21. Qwen3: Think Deeper, Act Faster | Qwen, accessed on March 7, 2026, https://qwenlm.github.io/blog/qwen3/
22. Benchmarked Qwen 3.5 small models (0.8B/2B/4B/9B) on few-shot learning — adding examples to 0.8B code tasks actually makes it worse : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rjbw0p/benchmarked_qwen_35_small_models_08b2b4b9b_on/
23. Qwen3.5 Model Series - Thinking On/OFF: Does it Matter? : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1riyfg2/qwen35_model_series_thinking_onoff_does_it_matter/
24. Qwen/Qwen3-4B-MLX-6bit - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3-4B-MLX-6bit
25. Qwen/Qwen3-0.6B - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3-0.6B
26. Qwen 3.5 Model: Switching Between Thinking and Non-Thinking #1559 - GitHub,

accessed on March 7, 2026, https://github.com/lmstudio-ai/lmstudio-bug-tracker/issues/1559
27. Alibaba Cloud Model Studio:Image and video understanding, accessed on March 7, 2026, https://www.alibabacloud.com/help/en/model-studio/vision
28. vLLM - Qwen, accessed on March 7, 2026, https://qwen.readthedocs.io/en/latest/deployment/vllm.html
29. Qwen/Qwen3.5-2B - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3.5-2B
30. Qwen3.5 Usage Guide - vLLM Recipes, accessed on March 7, 2026, https://docs.vllm.ai/projects/recipes/en/latest/Qwen/Qwen3.5.html
31. Eval bug: Qwen3.5-0.8B-GGUF throughput is significantly lower than expected · Issue #20072 · ggml-org/llama.cpp - GitHub, accessed on March 7, 2026, https://github.com/ggml-org/llama.cpp/issues/20072
32. Qwen3.5 4B: overthinking to say hello. : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rj8x1q/qwen35_4b_overthinking_to_say_hello/
33. Now in Foundry: Qwen3.5 Medium Model Series | Microsoft Community Hub, accessed on March 7, 2026, https://techcommunity.microsoft.com/blog/azure-ai-foundry-blog/now-in-foundry-qwen3-5-medium-model-series/4498640
34. Qwen3.5: Scaling 17B Activation for Expert Visual Coding Logic | by My Social - Medium, accessed on March 7, 2026, https://medium.com/aimonks/qwen3-5-scaling-17b-activation-for-expert-visual-coding-logic-b69da8f948f0
35. Qwen3-VL is the multimodal large language model series developed by Qwen team, Alibaba Cloud. · GitHub, accessed on March 7, 2026, https://github.com/QwenLM/Qwen3-VL
36. wangkanai/qwen3-vl-8b-instruct - Hugging Face, accessed on March 7, 2026, https://huggingface.co/wangkanai/qwen3-vl-8b-instruct
37. qwen3.5-397b-a17b Model by Qwen | NVIDIA NIM, accessed on March 7, 2026, https://build.nvidia.com/qwen/qwen3.5-397b-a17b/modelcard
38. Why do the Qwen 3.5 series benchmark better than Qwen 3 series? : r/LocalLLM - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLM/comments/1rm3u4r/why_do_the_qwen_35_series_benchmark_better_than/
39. Comprehensive Analysis of Qwen 3.5: Technical Inno... - U深研, accessed on March 7, 2026, https://unifuncs.com/s/xW70JGNc
40. qwen3.5:0.8b - Ollama, accessed on March 7, 2026, https://ollama.com/library/qwen3.5:0.8b
41. Technical AI model and multimodal updates | Yutori - Scouts, accessed on March 7, 2026, https://scouts.yutori.com/351b6c50-a7f6-42b0-ba33-cdf3436174b9
42. Qwen3-code-next at Q1 is beating Qwen3.5-35B-A3b at tool calling in my tests - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rmgt5v/qwen3codenext_at_q1

_is_beating_qwen3535ba3b_at/
43. unsloth/Qwen3.5-0.8B-GGUF - Hugging Face, accessed on March 7, 2026, https://huggingface.co/unsloth/Qwen3.5-0.8B-GGUF
44. Qwen3.5 - How to Run Locally Guide | Unsloth Documentation, accessed on March 7, 2026, https://unsloth.ai/docs/models/qwen3.5
45. It's time to turn off the annoying THINK MODE Qwen-3 | by Duke Wang | Medium, accessed on March 7, 2026, https://medium.com/@dukewillbe185/its-time-to-turn-off-the-annoying-think-mode-qwen-3-eefb7dedcadd
46. Qwen/Qwen3-VL-8B-Instruct - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct
47. qwen / qwen3-235b-a22b - NVIDIA, accessed on March 7, 2026, https://docs.api.nvidia.com/nim/reference/qwen-qwen3-235b-a22b
48. Qwen3.5 122B in 72GB VRAM (3x3090) is the best model available at this time — also it nails the "car wash test" - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rf2ulo/qwen35_122b_in_72gb_vram_3x3090_is_the_best_model/
49. LocalLlama - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/best/
50. You can use Qwen3.5 without thinking : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1re1b4a/you_can_use_qwen35_without_thinking/
51. How to switch Qwen 3.5 thinking on/off without reloading the model : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rhohqk/how_to_switch_qwen_35_thinking_onoff_without/
52. Is VLLM dynamic kwargs (qwen 3.5 thinking vs nonthinking) possible? - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rg5r4y/is_vllm_dynamic_kwargs_qwen_35_thinking_vs/
53. Qwen/Qwen3.5-27B - Hugging Face, accessed on March 7, 2026, https://huggingface.co/Qwen/Qwen3.5-27B
54. PSA: LM Studio's parser silently breaks Qwen3.5 tool calling and reasoning: a year of connected bug reports - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1riwhcf/psa_lm_studios_parser_silently_breaks_qwen35_tool/
55. Running Qwen3.5 27b dense with 170k context at 100+t/s decode and ~1500t/s prefill on 2x3090 (with 585t/s throughput for 8 simultaneous requests) : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1rianwb/running_qwen35_27b_dense_with_170k_context_at/
56. Gemma3 270m works great as a draft model in llama.cpp : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1mqj87e/gemma3_270m_works

_great_as_a_draft_model_in/
57. Repeat PP while using Qwen3.5 27b local with Claude Code : r/LocalLLaMA - Reddit, accessed on March 7, 2026, https://www.reddit.com/r/LocalLLaMA/comments/1ri6q8d/repeat_pp_while_using_qwen35_27b_local_with/
58. Issues | AINews, accessed on March 7, 2026, https://news.smol.ai/issues/
59. Qwen3.5 Released! Native Multimodality and Superior Performance | Proje Defteri, accessed on March 7, 2026, https://projedefteri.com/en/blog/qwen-3-5-review/
60. The Best Open-Source LLMs in 2026 - Bento, accessed on March 7, 2026, https://www.bentoml.com/blog/navigating-the-world-of-open-source-large-language-models
61. Why Multi-LLM Routing Is the Future of AI - API7.ai, accessed on March 7, 2026, https://api7.ai/blog/multi-llm-routing-ai-gateway-qwen
62. New Move by Qwen! Officially Open-Sources the Qwen3.5 Compact Model Series, Musk Expresses Views, accessed on March 7, 2026, https://news.futunn.com/en/post/69502828/new-move-by-qwen-officially-open-sources-the-qwen3-5
63. Alibaba Unveils New AI Model Series Qwen 3.5 - Binance, accessed on March 7, 2026, https://www.binance.com/en/square/post/03-03-2026-alibaba-unveils-new-ai-model-series-qwen-3-5-297427083420257